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METHOD AND APPARATUS FOR MANUFACTURING AN ELEVATOR GUIDE RAIL

Abstract

The present disclosure relates to a method and corresponding apparatus for manufacturing an elevator guide rail (**100**), comprising: providing (**310**) a hot-formed guide rail having, in a cross-section, a head portion (**110**) and a foot portion (**120**), and cold-forming (**320**) the provided guide rail through a die (**220**). The cold-forming is applied only to the head portion (**110**) and not to the foot portion (**120**), wherein the cold-formed head portion forms a tread (**112**) of the guide rail (**100**).

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a bypass continuation of prior Patent Cooperation Treaty (“PCT”) International Application No. PCT/EP2023/083261, filed Nov. 28, 2023, which claims priority to European Patent Application No. 22210251.9, filed Nov. 29, 2022. The entire disclosure of the documents cited in this paragraph is hereby incorporated herein by reference.

[0002] The present disclosure generally relates to the manufacturing of elevator guide rails. More particularly, the present disclosure relates to a method and apparatus for manufacturing an elevator guide rail where only a head portion of a hot-formed guide rail is further cold-formed.

[0003] Guide rails for a passenger lift or service lift provide for guiding the car of the lift or the counterweight of the lift along the moving direction of the car or the counterweight. The grades and quality as well as dimensional characteristics and tolerances of such guide rails are standardised, for example in norm ISO 7465:2007.

[0004] The norm distinguishes between cold-drawn rails (A type), machined rails (B type) and high precision machined rails (BE type). The tensile strength R_m of the raw material (steel) shall be at least 370 MPa (N/mm²) and not more than 520 MPa. During cold drawing, the material is pressed and, hence, its hardness and tensile strength increase. Thus, the norm recommends using steel grade E 235 B for A-type rails and steel grade E 275 B for B- and BE-type rails. Moreover, the norm defines a surface finish and roughness parameters, particularly for the guide rail blade, i.e., the surface on/along which a component of the car or counterweight moves.

[0005] However, machining the guide rails is time intensive so that, for the production of B-type and BE-type guide rails, this production step forms a bottleneck of the entire manufacturing. The machining is thus slow, increases production costs and requires large machinery, i.e., a large production space. In addition, in order to produce A-type rails as well as B- and/or BE-type rails, a large production space is also required.

[0006] It is thus an object of the present disclosure to provide a method and apparatus for manufacturing elevator guide rails in a more time and cost effective way.

[0007] According to a first aspect of the present disclosure, a method for manufacturing an elevator guide rail comprises providing a hot-formed guide rail having, in a cross-section, a head portion and a foot portion, and cold-forming the provided guide rail through a die, wherein said cold-forming is applied only to the head portion and not to the foot portion, wherein the cold-formed head portion forms a tread of the guide rail.

[0008] The cold-forming of the head portion avoids the time-consuming machining of the elevator guide rail, while the tread of the guide rail (i.e., the blade or surface on/along which a guiding component of the elevator car or counterweight glides/moves/runs/etc.) is formed in a single manufacturing step. For instance, while machining the guide rail may require minutes for each rail, the cold-forming can be achieved within seconds. Moreover, geometrical precision and a blank surface with low roughness is required for the tread (head portion) of the guide rail. The cold-forming provides such properties to the guide rail due to the compression of the material. Since only a portion of the guide rail is cold-formed, the cold-forming requires less power and is faster than cold-forming the entire surface of the guide rail as it is necessary for A-type guide rails.

[0009] Moreover, the norm suggests type A guide rails for rail sizes T45 to T90 and type B or BE guide rails for rail sizes T75 to T140. The method disclosed herein, on the other hand, allows manufacturing of any rail size with the same manufacturing steps. Thus, elevator guide rail manufacturing becomes less complex and the disclosed method is universally applicable.

[0010] In addition, since there is no machining of the guide rail, no material losses occur in the disclosed manufacturing method. This reduces the amount of raw material for producing the guide rails.

[0011] It is to be understood that any raw material can be employed and the present disclosure is not restricted to hot-formed guide rails. For instance, any hot-formed, e.g. hot-rolled or hot-drawn or hot-extruded or hot-forged, raw material can be employed for the production of a guide rail. Nevertheless, a hot-rolled guide rail can be provided with already precise dimensions, so that the further deformation, particularly during the cold-forming step of the disclosed method, is less time, energy and cost intensive.

[0012] In an implementation variant, said cold-forming can be adapted to achieve a deformation degree between 1.5% and 5%, preferably between 2.5% and 4%, and most preferably between 3.1% and 3.8%. In other words, the hot-formed raw guide rail is only slightly deformed at its head portion. For instance, for an average sized guide rail, the head thickness or width (when viewing the head portion in the cross-sectional direction of the guide rail, such as the size between two opposing tread surfaces of the head portion) can be reduced by approximately 0.3 to 0.6 mm, preferably 0.4 to 0.5 mm, and the head height can be reduced by approximately 0.4 to 0.8 mm, preferably 0.5 to 0.7 mm, due to the cold-forming.

[0013] Such small deformation degree allows cold-forming in a single step or in only a few cold-forming steps, such as two or three runs through corresponding dies sequentially having smaller cross-sectional openings.

[0014] Moreover, the deformation degree should not significantly fall under the lower limit of the above ranges since, otherwise, the desired surface properties may not be achieved. Specifically, having a deformation degree in the most preferred range can lead to a surface roughness required by the norm without further processing of the surface.

[0015] In another implementation variant, said providing the guide rail can comprise providing a guide rail of a steel grade between E235 and E275, preferably between E235 and E265. On the one hand, any recommended steel type can be provided to manufacture an elevator guide rail. Since the cold-forming compresses the material, its hardness and tensile strength increases. Thus, if employing E275 grade, the resulting elevator guide rail may exceed the maximum tensile strength of 520 MPa. Therefore, a steel grade lower than E275 can be employed and the resulting guide rail still fulfils the requirements of the norm.

[0016] Alternatively or additionally, said providing the guide rail can comprise providing a guide rail of a steel having a tensile strength between 260 MPa and 500 MPa, preferably between 350 MPa and 500 MPa, and most preferably between 450 MPa and 495 MPa. Such steel grade can be cold-formed without exceeding the maximum norm limit of 520 MPa.

[0017] Keeping the resulting tensile strength under the maximum norm limit of 520 MPa can be achieved by keeping the deformation degree in the above-mentioned range, for example, between 3.1% and 3.8%.

[0018] In yet another implementation variant, said providing the guide rail can comprise providing a guide rail having a T-shaped cross-section, wherein the head portion is formed at at least a portion of one leg of the T-shaped cross-section. The foot portion is formed by two flanges extending from the head portion.

[0019] Additionally or alternatively, said providing the guide rail can comprise providing a guide rail of one of the types T45 to T140 according to ISO 7465:2007.

[0020] Also additionally or alternatively, said providing the guide rail can comprise providing a guide rail having an L-shaped cross-section, wherein the head portion is formed at at least a portion of one leg of the L-shaped cross-section. The foot portion is formed by the other leg of the L-shaped cross-section.

[0021] Further additionally or alternatively, said providing the guide rail can comprise providing a guide rail having a cross-section without a distinct foot portion. As a mere example, such guide rail

can have a substantially rectangular cross-section. The head portion and a foot portion can be formed at opposite ends of the substantially rectangular cross-section. The cross-section may further have the same width over its entire height or may have at least two sections of different width. For instance, the foot portion can have a smaller cross-sectional width than the head portion of the substantially rectangular cross-section. Such type of guide rail allows mounting the guide rail, for example, in a clamping fixture, thereby reducing the space required for the mounting of the guide rail.

[0022] In general, the head portion and the foot portion respectively form parts of the guide rail having different functions. The foot portion is mainly used to mount the guide rail to a supporting structure of the elevator shaft. Furthermore, the foot portion can also be employed to connect the guide rail to another guide rail in a longitudinal direction thereof, for example, using a fish plate overlapping and mounted to both guide rails.

[0023] The head portion extends from the foot portion, such as into an open space of the elevator shaft when the guide rail is mounted in the elevator shaft. The head portion may be arranged at an angle to the foot portion (particularly when viewing the cross-section of the guide rail). As a mere example, the angle between the head portion and foot portion can be $90^{\circ} \pm 10^{\circ}$. According to another example, the angle between the head portion and foot portion can be 0° or 180° , i.e., head and foot portion together have a (substantially) rectangular cross-section. Thus, the head portion provides at least one surface, along which a guiding component of the elevator car or counterweight can glide, move, roll or run, when the elevator car or counterweight moves up and down the elevator shaft. This at least one surface hence forms a tread of the guide rail.

[0024] In any case, the cold-forming is applied to said at least one surface of the head portion, particularly at least one surface forming the tread of the guide rail. The cold-forming is not applied to a surface of the foot portion and/or the cold-forming may not be applied to a surface of the head portion not forming a tread (or blade) of the guide rail. This saves energy and time by reducing the overall level of deformation on the entire guide rail.

[0025] In a further implementation variant, said cold-forming comprises cold-rolling, cold-drawing, or a combination thereof. For instance, cold-rolling allows deformation of at least a portion of the guide rail with a reduced frictional component, so that less heat is generated during cold-forming. On the other hand, cold-drawing is achieved by providing a fixed (non-moving) die having a cross-sectional opening slightly smaller than the provided hot-formed guide rail. By drawing the guide rail through the die, at least a portion of the guide rail is deformed. It is to be understood that a combination of cold-rolling and cold-drawing is possible in any desired manner. For example, a particular surface of the guide rail, such as the tread or blade surface(s), can be cold-rolled, while a top of the head portion is cold-drawn, or vice versa. In any case, a portion of the guide rail becomes smaller in at least one cross-sectional direction.

[0026] In an implementation variant, the method can further comprise heating the cold-formed guide rail to anneal at least the head portion of the guide rail. For example, if the raw steel already has a high tensile strength, cold-forming may lead to a tensile strength exceeding the maximum of 520 MPa. By annealing at least the head portion of the guide rail, the tensile strength and other mechanical properties induced by the cold-forming can be reduced. As an example only, the heating can be achieved by induction heating of at least the head portion of the cold-formed guide rail.

[0027] According to a second aspect of the present disclosure, an elevator guide rail comprises a head portion forming a tread of the guide rail, and a foot portion, wherein the guide rail is manufactured according to (or “obtained by”) the method of the first aspect or at least one of its variants.

[0028] The elevator guide rail, which is only partially cold-formed, may have particular material properties in a surface region of the cold-formed head portion. For instance, as the material of the guide rail is partially work-hardened due to the cold-forming process, it will have an increased

tensile strength. For instance, the cold-forming can increase the tensile strength of the head portion up to 100 MPa compared to the (non-formed) foot portion. In addition, a smoother surface, i.e., a smaller roughness parameter value, may be achieved during the cold-forming.

[0029] Thus, the elevator guide rail according to the second aspect can be distinguished from guide rails produced by another (e.g., conventional) manufacturing process by comparing the tensile strength of the head portion and the foot portion and determining whether the tensile strength of the head portion is higher. Likewise, the surface roughness can be smaller (smoother) at the head portion compared to the foot portion.

[0030] In addition, a microscopic analysis of the elevator guide rail according to the second aspect may show an elongated microstructure of the steel, particularly in the surface region of the cold-formed (part of the) head portion. Thus, the elevator guide rail according to the second aspect can be distinguished from guide rails produced by another (e.g., conventional) process by comparing the microstructure at the head portion, for instance at a tread, and at the foot portion, for example.

[0031] According to a third aspect to of the present disclosure, an apparatus for manufacturing an elevator guide rail comprises a receiving component configured to receive a hot-formed guide rail having, in a cross-section, a head portion and a foot portion, and a die configured to cold-form only the head portion and not the foot portion of the provided guide rail, wherein the cold-formed head portion forms a tread of the guide rail. The apparatus further comprises a moving component configured to move the guide rail through the die. For instance, the moving component can pull the guide rail through the die.

[0032] In an implementation variant, the die can be implemented by one or more propelled or non-propelled rollers that cold-roll at least a portion of the guide rail. For instance, a pair of vertical rollers and a pair of horizontal rollers can be employed. Alternatively or additionally, the die may function as a cold-drawing component by providing a fixed die having a cross-sectional opening slightly smaller than the provided hot-formed guide rail. It is to be understood that a combination of rollers and fixed die can be achieved in any desired form, so that portions of the guide rail are cold-rolled and other portions are cold-drawn. In any case, a portion of the guide rail becomes smaller in at least one cross-sectional direction.

[0033] In another implementation variant, the die can comprise a plurality of partial dies, each of which cold-forms a surface of the head portion. For instance, each partial die may cold-form a surface of the head portion, so that the number of deformed surfaces corresponds to the number of partial dies. Alternatively, a partial die may deform more than one surface of the guide rail, such as an L-shaped partial die.

[0034] As an example only, the plurality of partial dies can comprise at least one roller for cold-rolling the corresponding portion of the head portion of the guide rail. Optionally, the remaining partial die or dies of the plurality of partial dies is/are one or more fixed dies.

[0035] In yet another implementation variant, the die can comprise at least one guiding block, each of which is configured to support the guide rail in the die. In other words, the die comprises at least one portion that does not deform a corresponding surface of the guide rail, but allows gliding or rolling of the guide rail along the guiding block.

[0036] In a further implementation variant, the apparatus can further comprise a heating component configured to heat at least a portion of the cold-formed guide rail. Such heating component can be implemented in-line in the production process, such as close to or directly after the die (in the moving direction of the guide rail through the apparatus).

[0037] This particularly allows annealing at least the head portion of the guide rail, in order to reduce particular mechanical properties of the cold-formed guide rail.

[0038] As an example only, the heating component can be or can comprise an induction heating component or induction heater configured to increase the temperature of at least the head portion of the cold-formed guide rail.

[0039] The present disclosure is not restricted to the described aspects and variants in the described

form and order. Specifically, the description of aspects and variants is not to be understood as a specific limiting grouping of features. It is to be understood that the present disclosure also covers combinations of the aspects and variants not explicitly described. Thus, each variant or optional feature can be combined with any other aspect, variant, optional feature or even combinations thereof.

Description

[0040] In the following, the present disclosure will further be described with reference to exemplary embodiments illustrated in the figures, in which:

[0041] FIGS. **1A** and **1B** schematically illustrate cross-sections of an exemplary elevator guide rails;

[0042] FIG. **2** schematically illustrates a perspective view of an exemplary apparatus for manufacturing an elevator guide rail;

[0043] FIG. **3** schematically illustrates a plurality of partial dies;

[0044] FIG. **4** schematically illustrates a side view of an exemplary apparatus for manufacturing an elevator guide rail; and

[0045] FIG. **5** illustrates a flow diagram of a method for manufacturing an elevator guide rail.

[0046] In the following description, for purposes of explanation and not limitation, specific details are set forth in order to provide a thorough understanding of the present disclosure. It will be apparent to one skilled in the art that the present disclosure may be practiced in other implementations that depart from these specific details.

[0047] FIG. **1A** schematically illustrates a cross section of an exemplary elevator guide rail **100**.

Such guide rail **100** may have a T-shaped cross-section forming a head portion **110** and a foot portion **120**, which is only one example of a guide rail. The head and foot portion is **110**, **120** may also be referred to as head section and foot section, respectively. Particularly, the foot portion **120** may be formed by two flanges extending from the head portion **110**. The head portion **110** can form or comprise a tread **112** (also referred to as a blade or gliding surface or guiding surface) of the guide rail **100**. A car or counterweight of the elevator (not illustrated) usually has a component connected thereto that runs along the guide rail **100**, so that the car or counterweight is guided along its path up and down the elevator shaft. Such component may include rollers or gliders that contact the tread **112** of the head portion **110**.

[0048] Optionally, the head portion **110** can further be separated into a section forming the tread **112** of the guide rail **100** and an intermediate section **115**. The intermediate section **115**, particularly a length of the intermediate section **115**, may hence solely facilitate placing the tread **112** in the right position in relation to the foot portion **120**, i.e., in the right position within the elevator shaft.

[0049] FIG. **1B** schematically illustrates a cross-section of another exemplary elevator guide rail **100**. Such guide rail **100** may have a substantially L-shaped cross-section forming a head portion **110** and a foot portion **120**, which is also only one example of a guide rail. Particularly, in this example, the foot portion **120** extends from the head portion **110** in one direction at an angle with respect to an extent of the head portion **110**. This allows reducing the space required for the mounting of the guide rail **100** compared to a guide rail **100** having a T-shaped cross-section (as in FIG. **1A**). The head portion **110** can form or comprise a tread **112**, as in the example of FIG. **1A**. The detailed description of such tread **112** will hence be omitted to avoid redundant explanations.

[0050] It is to be noted that, in FIGS. **1A** and **1B**, the head portion **110** and the foot portion **120** as well as the intermediate section **115** are illustrated as having a different width. While this may reflect the actual form of such guide rail **100**, FIGS. **1A** and **1B** use this illustration also to distinguish between the tread **112** of the head portion **110** and the foot portion **120**. A guide rail **100**, however, is not restricted to such cross-sectional shape. For instance, a guide rail **100** may

have the same cross-sectional width over the entire height (of the head portion **110**) and over the entire extent of the one or two flanges of the foot portion **120**.

[0051] FIG. **2** schematically illustrates a perspective view of an exemplary apparatus **200** for manufacturing an elevator guide rail **100**. The apparatus **200** comprises a receiving component **210** configured to receive a guide rail **100**. The receiving component **210** can simply be implemented as rollers on which the guide rail **100** rolls. It is to be understood that the receiving component **210** can have any shape and form for controlling a movement of the guide rail **100** along its longitudinal axis (X-axis).

[0052] The guide rail **100** can comprise a head portion **110** and a foot portion **120** as described with respect to FIG. **1A** or **1B**. The apparatus **200** further comprises a die **220** configured to cold-form a portion of the guide rail **100**, particularly at least a part of the head portion **110**. This part of the head portion **110** forms a tread **112** of the guide rail **100**. Such tread **112** requires particular mechanical properties and surface parameters, in order to facilitate guiding the car or counterweight of the elevator with reduced friction and small tolerances. In order to achieve the required mechanical properties and surface parameters, the die **220** cold-forms this part of the head portion **110**. As is exemplary illustrated in FIG. **2**, the die **220** comprises an upper section **221** acting in a Z-axis direction on a top surface of the head portion **110**, and two side sections **222**, **223** acting in a Y-axis direction and opposite to one another. The side sections **222**, **223** form the tread **112** on each side of the guide rail **100**. The top surface of the head portion **110** can also form a tread, i.e., a surface having a low roughness for good gliding and guiding capabilities of the guide rail **100**.

[0053] The die **220** can comprise a plurality of partial dies, each of which cold-forms a surface of the head portion **110**. It is to be understood that the die **220** may comprise less partial dies than illustrated in the figures. For instance, the upper section **221** of the die **220** can be omitted, if the corresponding surface of the guide rail **100** does not need to be cold-formed. With additional reference to FIG. **3**, the partial dies **221** to **223** can comprise one or more rollers for cold-rolling the corresponding part of the head portion **110**. As a mere example, side sections **222**, **223** may be implemented as horizontal rollers cold-forming the tread **112** of the guide rail **100**. The upper section **221** can be a fixed die cold-forming the top surface of the guide rail **100**.

[0054] While the die **220** can comprise only these sections **221** to **223** cold-forming the head portion **110**, the die **220** can optionally include further die sections. For instance, there can be a bottom block **240**, which is configured to support the guide rail **100** in the die **220**. Such bottom block **240** can form a support for the guide rail **100**, particularly against the forces induced by cold-forming upper section **221** of die **220**. Furthermore, additional side sections **232**, **233** may be present, which guide the guide rail **100** through the die **220**, but without cold-forming the corresponding part of the guide rail **100**. Such side sections **232**, **233** and/or bottom block **240** may have a larger size, so that an area supporting the guide rail **100** is much larger than a contact surface between the cold-forming portions of the die **220** and the guide rail **100**. This avoids cold-forming the guide rail **100** in areas not forming the tread **112**.

[0055] It is to be understood that the size, form and location of the partial dies **221** to **223** and guiding blocks **232**, **233**, **240** are for illustrative purposes only. These parts of the die **220** can generally have any form and shape necessary to cold-form (at least a part of) the head portion **110** of the guide rail **100**, while leaving the remaining parts of the guide rail **100** unaltered.

[0056] Again with reference to FIG. **2**, the guide rail **100** may have through holes **150** at at least one end thereof. Such through holes **150** may allow connecting the guide rail **100** to another guide rail **100** (not illustrated) disposed in a longitudinal direction of the first guide rail **100**. For instance, a fish plate may be mounted underneath the guide rail **100** using fasteners inserted into the through holes **150**.

[0057] Such through holes **150** may further be used to couple the guide rail **100** to a moving component **250** of the apparatus **200** that is configured to move the guide rail **100** through the die

220. The moving component **250**, however, can be coupled to the guide rail **100** in any other manner, such as welded to the guide rail **100**, mounted to the head portion **110** and/or the intermediate section **115**, for example.

[0058] FIG. **4** schematically illustrates a side view of the apparatus **200**. Components that are identical to those already illustrated in and described with respect to FIGS. **2** and **3** have been indicated with the same reference signs and their description will be omitted to avoid redundant explanations. The moving component **250** can be arranged at a longitudinal end of the guide rail **100** and moves the guide rail **100** in the longitudinal direction thereof (the X-axis).

[0059] After cold-forming the tread **112** in the die **220**, the mechanical properties of the guide rail **100** have changed. Particularly, a tensile strength of the head portion **110** may have increased. In order to anneal at least the head portion **110**, a heating component **280** may increase the temperature of the head portion **110**. For example, the heating component **280** can be an induction heating component, which facilitates heating only a portion of the guide rail **100**, such as the head portion **110** and/or the tread **112**. The heating component **280** may further be configured to only heat a surface of the guide rail **100**. This not only saves energy, but allows maintaining the mechanical properties of the cold-formed head portion **110** as much as possible.

[0060] FIG. **5** illustrates a flow diagram of a method for manufacturing an elevator guide rail **100**, such as the guide rail **100** illustrated in FIGS. **1** to **4**. The method starts, in step **310**, by providing a hot formed (e.g., hot-rolled) guide rail **100** having a head portion **110** and a foot portion **120**, such as guide rail **100** illustrated in FIGS. **1** and **2**.

[0061] The hot-formed guide rail **100** may be of a steel grade between E 235 and E 275, preferably between E 235 and E 265. Such steel grade can be cold-formed, while keeping the mechanical properties within limits defined by the norm for elevator guide rails, such as ISO 7465:2007.

[0062] Alternatively or additionally, in step **310**, there is provided a guide rail of a steel having a tensile strength between 260 MPa and 500 MPa, preferably between 350 MPa and 500 MPa, and most preferably between 450 MPa and 495 MPa. Even if the tensile strength increases due to cold-forming, the resulting tensile strength, particularly in the area of the head portion **110**, is below a limit of 520 MPa specified in the above-mentioned norm.

[0063] The chemical composition of the raw guide rail **100**, according to a mere example, is given in table 1 below.

TABLE-US-00001

TABLE 1 exemplary chemical composition of the steel for the guide rail 100				
C	Mn	S	Al	
0.075	0.368	0.0076	0.0034	

[0064] In a following step **320**, only a head portion **110** of the guide rail **100** is cold-formed. Such cold-forming, for example in a die **220**, changes some of the mechanical properties of the guide rail **100**. This is illustrated in below table 2 showing some test results performed on a guide rail **100** manufactured according to the disclosed method. The samples for testing the mechanical properties of the guide rail **100** have been taken from the head portion **110** and from each side of the foot portion **120**, i.e., one sample from each flange forming the foot portion **120**.

TABLE-US-00002

TABLE 2 mechanical properties of samples of the guide rail 100				
	Tensile	yield strength	yield Brinell	Elongation Rm, RP0.2, ratio hardness at fracture
Sample	MPa	MPa	RP0.2/Rm	HB A5, %
Head portion	510	443	0.87	152
Foot portion L	493	442	0.9	147
Foot portion R	464	389	0.84	135

[0065] As is derivable from above Table 2, the cold-forming of the head portion **110** increases the tensile strength, but keeps it below 520 MPa, which is a maximum limit required by the applicable norm ISO 7465:2007.

[0066] In order to achieve such mechanical properties, the cold-forming in step **320** may provide a deformation degree between 1.5% and 5%, preferably between 2.5% and 4%, and most preferably between 3.1% and 3.8%. The upper limit of such deformation degree range prevents the guide rail **100** from having a higher tensile strength, while the lower limit of such deformation degree range achieves the desired surface structure (surface parameters, such as roughness).

[0067] In a further step **330**, the cold-formed guide rail may be heated to anneal at least the head portion **110** of the guide rail **100**. This heating of the head portion **100** reduces the tensile strength and further allows straightening the guide rail **100**, which may have been bent due to the cold-forming of only the head portion **110**. The heating may be applied by an induction heating component, which is easy to install in-line with the die **220** and allows specific heating of the guide rail **100** being made of a ferrous metal.

[0068] The following numbered statements describe some various embodiments of the present invention.

[0069] Statement #1. A method for manufacturing an elevator guide rail (**100**), the method comprising: [0070] providing (**310**) a hot-formed guide rail having, in a cross-section, a head portion (**110**) and a foot portion (**120**); and [0071] cold-forming (**320**) the provided guide rail through a die (**220**), [0072] characterised in that [0073] said cold-forming is applied only to the head portion (**110**) and not to the foot portion (**120**), wherein the cold-formed head portion forms a tread (**112**) of the guide rail (**100**).

[0074] Statement #2. The method of Statement #1, wherein said cold-forming is adapted to achieve a deformation degree between 1.5% and 5%, preferably between 2.5% and 4%, and most preferably between 3.1% and 3.8%.

[0075] Statement #3. The method of Statement #1 or Statement #2, wherein said providing (**310**) the guide rail (**100**) comprises providing a guide rail of a steel grade between E 235 and E 275, preferably between E 235 and E 265, or providing a guide rail of a steel having a tensile strength between 260 MPa and 500 MPa, preferably between 350 MPa and 500 MPa, and most preferably between 450 MPa and 495 MPa.

[0076] Statement #4. The method of one of Statement #1 to Statement #3, wherein said providing (**310**) the guide rail (**100**) comprises providing a guide rail having a T-shaped cross-section or an L-shaped cross-section or a rectangular cross-section, wherein the head portion (**110**) is formed at at least a portion of one leg of the T-shaped or L-shaped cross-section or a portion of the rectangular cross-section, and the foot portion (**120**) is formed by the remaining portion of the cross-section, and/or [0077] wherein said providing the guide rail (**100**) comprises providing a guide rail type T45 to T140 according to ISO 7465:2007.

[0078] Statement #5. The method of one of Statement #1 to Statement #4, wherein said cold-forming (**320**) comprises cold-rolling, cold-drawing, or a combination thereof.

[0079] Statement #6. The method of one of Statement #1 to Statement #5, further comprising:

[0080] heating (**330**) the cold-formed guide rail (**100**) to anneal at least the head portion (**110**) of the guide rail (**100**), preferably by induction heating of at least the head portion (**110**) of the guide rail (**100**).

[0081] Statement #7. An elevator guide rail (**100**), comprising: [0082] a head portion (**110**) forming a tread (**112**) of the guide rail (**100**); and [0083] a foot portion (**120**), [0084] wherein the guide rail (**100**) is manufactured according to the method of any one of the preceding Statement Nos.

[0085] Statement #8. An apparatus (**200**) for manufacturing an elevator guide rail (**100**), the apparatus comprising: [0086] a receiving component (**210**) configured to receive a hot-formed guide rail (**100**) having, in a cross-section, a head portion (**110**) and a foot portion (**120**); [0087] a die (**220**) configured to cold-form only the head portion (**110**) and not the foot portion (**120**) of the provided guide rail (**100**), wherein the cold-formed head portion (**110**) forms a tread (**112**) of the guide rail (**100**); and [0088] a moving component (**250**) configured to move the guide rail (**100**) through the die (**220**).

[0089] Statement #9. The apparatus (**200**) of Statement #8, wherein the die (**220**) comprises a plurality of partial dies (**221** to **223**), each of which cold-forms a surface of the head portion (**110**).

[0090] Statement #10. The apparatus (**200**) of Statement #8 or Statement #9, wherein the plurality of partial dies (**221** to **223**) comprises at least one roller for cold-rolling the corresponding portion of the head portion (**110**).

[0091] Statement #11. The apparatus (200) of Statement #10, wherein the remaining partial dies (221 to 223) of the plurality of partial dies are fixed dies.

[0092] Statement #12. The apparatus (200) of any of Statement #8 to Statement #11, wherein the die (220) comprises at least one guiding block (232, 233, 240), each of which is configured to support the guide rail (100) in the die (220).

[0093] Statement #13. The apparatus (200) of one of Statement #8 to Statement #12, further comprising: [0094] a heating component (280) configured to heat at least a portion of the cold-formed guide rail (100), preferably configured to anneal at least the head portion (110) of the guide rail (100).

[0095] Statement #14. The apparatus (200) of Statement #13, wherein the heating component (280) is an induction heating component.

[0096] It is believed that the advantages of the technique presented herein will be fully understood from the foregoing description, and it will be apparent that various changes may be made in the form, constructions and arrangement of the exemplary aspects thereof without departing from the scope of the disclosure or without sacrificing all of its advantageous effects. Because the technique presented herein can be varied in many ways, it will be recognized that the disclosure should be limited only by the scope of the claims that follow.

Claims

1. A method for manufacturing an elevator guide rail, the method comprising: providing a hot-formed guide rail having, in a cross-section, a head portion and a foot portion; and cold-forming the provided guide rail through a die, characterised in that said cold-forming is applied only to the head portion and not to the foot portion, wherein the cold-formed head portion forms a tread of the guide rail.
2. The method of claim 1, wherein said cold-forming is adapted to achieve a deformation degree between 1.5% and 5%, preferably between 2.5% and 4%, and most preferably between 3.1% and 3.8%.
3. The method of claim 1, wherein said providing the guide rail comprises providing a guide rail of a steel grade between E 235 and E 275, preferably between E 235 and E 265, or providing a guide rail of a steel having a tensile strength between 260 MPa and 500 MPa, preferably between 350 MPa and 500 MPa, and most preferably between 450 MPa and 495 MPa.
4. The method of claim 1, wherein said providing the guide rail comprises providing a guide rail having a T-shaped cross-section or an L-shaped cross-section or a rectangular cross-section, wherein the head portion is formed at at least a portion of one leg of the T-shaped or L-shaped cross-section or a portion of the rectangular cross-section, and the foot portion is formed by the remaining portion of the cross-section, and/or wherein said providing the guide rail comprises providing a guide rail type T45 to T140 according to ISO 7465:2007.
5. The method of claim 1, wherein said cold-forming comprises cold-rolling, cold-drawing, or a combination thereof.
6. The method of claim 1, further comprising: heating the cold-formed guide rail to anneal at least the head portion of the guide rail, preferably by induction heating of at least the head portion of the guide rail.
7. An elevator guide rail, comprising: a head portion forming a tread of the guide rail; and a foot portion, wherein the guide rail is manufactured according to the method of claim 1.
8. An apparatus for manufacturing an elevator guide rail, the apparatus comprising: a receiving component configured to receive a hot-formed guide rail having, in a cross-section, a head portion and a foot portion; a die configured to cold-form only the head portion and not the foot portion of the provided guide rail, wherein the cold-formed head portion forms a tread of the guide rail; and a moving component configured to move the guide rail through the die.

- 9.** The apparatus of claim 8, wherein the die comprises a plurality of partial dies, each of which cold-forms a surface of the head portion.
- 10.** The apparatus of claim 9, wherein the plurality of partial dies comprises at least one roller for cold-rolling the corresponding portion of the head portion.
- 11.** The apparatus of claim 10, wherein the remaining partial dies of the plurality of partial dies are fixed dies.
- 12.** The apparatus of claim 8, wherein the die comprises at least one guiding block, each of which is configured to support the guide rail in the die.
- 13.** The apparatus of claim 8, further comprising: a heating component configured to heat at least a portion of the cold-formed guide rail, preferably configured to anneal at least the head portion of the guide rail.
- 14.** The apparatus of claim 13, wherein the heating component is an induction heating component.
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